SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour
QUESTIONS: 4 **Total: Marks:40**

Annexure A

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Code of Conduct:

I AGNEESHWARAN B / 192524508 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AGNEESHWARAN B

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

ANSWER SHEET – ANNEXURE A

Industry Problem: Diabetes Diagnosis and Personalised Treatment Recommendation

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

(a) Inductive Learning Approach:
A supervised inductive learning approach is suitable because the patient records contain input features and a known target outcome such as Diabetes (Yes/No). The model learns patterns from labelled training examples and predicts diabetes for new patients.

Target variable: Diabetes diagnosis (Diabetic / Non-diabetic).

Key features:
1. Glucose level – a major indicator of diabetes risk because high glucose is strongly associated with diabetes.
2. BMI – higher BMI can indicate increased metabolic risk and is useful for classification.
3. Blood pressure – abnormal blood pressure can be associated with diabetes and related complications.

Other useful features include age, cholesterol and family history.

(b) Class Imbalance:
If non-diabetic cases are much more numerous, a classifier may become biased toward the majority class. I would use SMOTE on the training data because it creates synthetic minority-class samples without simply duplicating existing cases. It can improve recall for diabetic patients. SMOTE must be applied only to the training set to avoid data leakage. Class weights can also be used as an alternative.

(c) 70:30 Split and Preprocessing:
A 70% training and 30% testing split provides enough data to learn the model while retaining a sufficiently large unseen test set for evaluation. In a medical application, stratified splitting should be used so that diabetic and non-diabetic proportions are preserved.

Preprocessing:
- Handle missing values using suitable imputation.
- Encode categorical variables such as family history.
- Normalise/standardise numerical variables when required by the chosen model.
- Apply SMOTE only after splitting, on the training data.

These steps improve data quality, prevent leakage and make model training more reliable.

TASK 2: DECISION TREE FOR DIAGNOSIS

(a) Decision Tree:
A Decision Tree classifier can be trained using features such as glucose, BMI, blood pressure, age, cholesterol and family history. The root node should be selected using the feature having the highest Information Gain or the largest Gini impurity reduction.

Illustrative first three levels:

```
Level 0: Glucose
├── Low → Non-diabetic
└── High → Level 1: BMI
    ├── Normal → Level 2: Age
    │   ├── Younger → Non-diabetic
    │   └── Older → Higher-risk
    └── High → Level 2: Family History
        ├── No → Moderate-risk
        └── Yes → Diabetic
```

The exact root and branches must be obtained from the trained dataset using the calculated Information Gain/Gini values; the above is an illustrative structure.

(b) Evaluation:
The model should be evaluated using Accuracy, Precision, Recall and F1-Score.

Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 = 2 × Precision × Recall / (Precision + Recall)

False Negatives are patients who actually have diabetes but are predicted as non-diabetic. In this medical use-case, reducing False Negatives is especially important because missed cases may delay diagnosis and treatment. A suitable approach is to tune the decision threshold, use class weights or SMOTE, and optimise for recall while monitoring precision.

(c) Post-pruning:
Pre-pruning limits tree complexity during training using settings such as maximum depth or minimum samples per split. Post-pruning first grows a larger tree and then removes branches using cost-complexity pruning.

A comparison should report the measured test accuracy of both models. The preferred clinical model should balance accuracy with recall, interpretability and generalisation. A slightly lower accuracy model may be preferable if it substantially reduces false negatives and avoids overfitting.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

(a) Statistical Model:
Logistic Regression is suitable because diabetes diagnosis is a binary outcome. It estimates the probability that a patient belongs to the diabetic class.

The Decision Tree and Logistic Regression should be compared using:
• Accuracy
• AUC-ROC

AUC-ROC measures how well the model separates diabetic and non-diabetic patients across classification thresholds.

A statistical model is preferable when interpretability, stable probability estimates and understanding of feature effects are important. Logistic Regression is also useful when the relationship between predictors and log-odds is reasonably suitable for a linear model.

No actual accuracy/AUC values are supplied in the assessment sheet, so they should be calculated from the given dataset rather than invented.

(b) Feature Importance:
For the Decision Tree, importance can be obtained from impurity-based feature importance. For Logistic Regression, the magnitude of standardised coefficients can be examined.

The top three predictors from each model should be listed and compared. If the same features appear in both models, they provide stronger evidence that those variables are important. Differences can occur because the two models learn relationships differently: trees capture nonlinear interactions and thresholds, while Logistic Regression models additive effects on the log-odds scale.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

(a) MDP Formulation:
States: Patient health stages, for example:
S1 = High-risk/poor condition
S2 = Moderate-risk condition
S3 = Stable/improving condition
S4 = Good/controlled condition

Actions:
A1 = Diet
A2 = Exercise
A3 = Medication
A4 = Monitor

Reward function:
Give a positive reward when the patient's health state improves, a smaller reward for maintaining a stable state, and a negative reward for deterioration. Medication-related actions should only be permitted according to appropriate clinical rules.

Transition dynamics:
After an action, the patient moves probabilistically to another health state depending on the patient's current state and the selected treatment action. The transition model represents how treatment affects health over time.

(b) Q-Learning:
The Q-learning update is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[r + \gamma \max Q(s',a') - Q(s,a)]$$

where α is the learning rate, γ is the discount factor, r is the reward, s is the current state, a is the action and s' is the next state.

Illustrative example with $\alpha = 0.5$ and $\gamma = 0.9$:

Iteration 1:
$Q(S1,Diet) = 0$
Reward = 5, next state's maximum Q = 2
New $Q = 0 + 0.5[5 + 0.9(2) - 0] = 3.40$

$Q(S1,Exercise) = 0$
Reward = 4, next max Q = 3
New $Q = 0 + 0.5[4 + 0.9(3)] = 3.35$

$Q(S2,Monitor) = 0$
Reward = 2, next max Q = 3
New $Q = 0 + 0.5[2 + 0.9(3)] = 2.35$

Iteration 2, using illustrative next-state values:

$Q(S1,Diet) = 3.40$
Reward = 5, next max Q = 3
New $Q = 3.40 + 0.5[5 + 0.9(3) - 3.40] = 5.15$

$Q(S1,Exercise) = 3.35$
Reward = 4, next max Q = 3
New $Q = 3.35 + 0.5[4 + 0.9(3) - 3.35] = 4.00$

$Q(S2,Monitor) = 2.35$
Reward = 2, next max Q = 3
New $Q = 2.35 + 0.5[2 + 0.9(3) - 2.35] = 2.675$

The Q-learning agent should be trained for at least 5 patient episodes as required. A suitable convergence criterion is that the maximum absolute change in Q-values between successive iterations remains below a small threshold ε for several consecutive iterations.

(c) Final Policy and Ethics:
The final policy is obtained by selecting the action with the highest Q-value for each state:

$\pi(s) = \text{argmax}_a Q(s,a)$

Illustratively, if Diet has the highest Q-value in S1, the policy selects Diet in S1. The actual final policy must be obtained from the trained Q-table.

Ethical considerations:
- Patient safety – the agent must not independently make unsafe treatment decisions.
- Bias – training data should represent relevant patient populations and should be checked for systematic performance differences.
- Explainability – recommendations should be understandable to clinicians.
- Human oversight – a qualified healthcare professional should review recommendations.
- Privacy – patient data must be protected.
- Clinical validation – the system must be tested and monitored before deployment.
- Medication safety – medication recommendations require strict clinical constraints and should not be based only on learned rewards.